

8 (a)(i) Define *gravitational field strength*.

..... The gravitational field strength at a point is the gravitational force exerted
 per unit mass when a small test mass is placed at that point in a
 gravitational field.

..... [1]

(ii) Derive, from Newton's Law of Gravitation and (a)(i) above, the equation for the gravitational field strength of a point mass of mass M_1 , placed in a gravitational field generated by a mass M_2 and at a distance of r from M_1 . [2]

Since $g = F/M_1$
 and $F = GM_1M_2/r^2$ [Both equations to get 1]

Therefore, $g = GM_1M_2/r^2 / M_1$
 $= GM_2/r^2$ [1]

(b) Information related to the Earth and the Moon is given below:

$$\frac{\text{Radius of Earth}}{\text{Radius of Moon}} = 3.7$$

$$\frac{\text{Mass of Earth}}{\text{Mass of Moon}} = 81$$

The center-to-center distance of the Moon from the Earth is 3.84×10^8 m and the gravitational field strength due to the Earth at its surface is 9.8 N kg^{-1} .

(i) Using these data, calculate the gravitational field strength due to the Moon at its surface.

Since $g \propto M/r$,
 $\frac{g_M}{g_E} = \left(\frac{M_M}{M_E}\right) \left(\frac{r_E}{r_M}\right)^2$ [M1]
 $= (1/81)(3.7)^2$
 $= 0.169$

Therefore, g due to Moon $= 0.169 \times 9.81$
 $= 1.66 \text{ N kg}^{-1}$ [A1]

gravitational field strength = N kg^{-1} [3]

- (ii) There is a point on the line between the Earth and the Moon at which their combined gravitational field strength is zero.

Calculate the distance between this point and the centre of the Earth.

$$\frac{GM_E}{r^2} = \frac{GM_M}{(3.84 \times 10^8 - r)^2}$$

$$81 = \frac{r^4}{(3.84 \times 10^8 - r)^2} [1]$$

$$r = 3.46 \times 10^8 \text{ m} [1]$$

distance = m [2]

- (iii) The Moon orbits around the Earth with a period of 27.3 days.

1. Calculate the angular speed of the Moon.

$$\omega = 2\pi/T$$

$$= 2\pi/(27.3 \times 24 \times 60 \times 60)$$

$$= 2.66 \times 10^{-6} \text{ rad s}^{-1} [1]$$

angular speed = rad s⁻¹ [1]

2. Calculate the mass of the Earth.

Gravitational acceleration provides centripetal acceleration.

$$GM/r^2 = r\omega^2$$

$$M = r^3\omega^2/G [M1]$$

$$= (3.84 \times 10^8)^3 \times (2.66 \times 10^{-6})^2 / 6.67 \times 10^{-11}$$

$$= 6.0 \times 10^{24} [A1]$$

..... kg [2]

3. Determine the gravitational force between the Earth and the Moon.

Mass of Moon = 1/81 Mass of Earth

$$F = GM_E M_M / r^2$$

$$= [6.67 \times 10^{-11} (6.0 \times 10^{24})^2 / 81] / (3.84 \times 10^8)^2 [1]$$

$$= 2.01 \times 10^{20} [1]$$

gravitational force = N [2]

4. Tidal action on the Earth's surface causes the radius of the orbit of the Moon to increase by 4.0 cm each year. Using your answer in **(b)(iii)(3)**, determine the change, in one year, of the gravitational potential energy of the Moon.

$$\begin{aligned} \text{Change in potential energy} \\ &= \text{Work done by external force} \\ &= 2.01 \times 10^{20} \times 0.04 \text{ [1]} \\ &= 8.04 \times 10^{18} \text{ J [1]} \end{aligned}$$

change in potential energy = J [2]

- (c) (i) Explain, by considering the respective field force, why gravitational potential energy is negative whereas electric potential energy can be positive or negative.

Gravitational forces are always attractive, whereas electric forces can be attractive or repulsive. [1]

.....[1]

- (ii) The Earth may be assumed to be an isolated sphere of radius 6.4×10^3 km with its mass of 6.0×10^{24} kg concentrated at its centre. A 2.0 kg mass is projected vertically from the surface of the Earth so that it reaches a maximum altitude of 1.3×10^4 km.

Calculate, for this mass,

1. the change in gravitational potential energy

$$\begin{aligned} \text{Change in GPE} \\ &= -GMm/r_2 - (-GMm/r_1) \\ &= GMm (1/r_1 - 1/r_2) \\ &= 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 2.0 (1/6.4 \times 10^6 - 1/(6.4 \times 10^6 + 1.3 \times 10^7)) \text{ [M1]} \\ &= 8.38 \times 10^7 \text{ J [A1]} \end{aligned}$$

change in gravitational potential energy = J [2]

2. the speed of projection from the Earth's surface, assuming air resistance is negligible.

$$\begin{aligned} \text{By COE, Loss in KE} &= \text{Gain in GPE} \\ \frac{1}{2} m v^2 &= 8.38 \times 10^7 \\ \frac{1}{2} (2) v^2 &= 8.38 \times 10^7 \text{ [M1]} \\ v &= 9150 \text{ m s}^{-1} \text{ [A1]} \end{aligned}$$

speed = m s⁻¹ [2]

9 (a) By reference to energy transfers distinguish between electromotive force (e.m.f.) and